DEM 5283/7283 Course Topics

Statistics for Demographic Data II, Spring 2027

Course Description:

This course represents an in-depth coverage of the general linear model framework, including alternative logistic regression models, count-data regression and multi-level modeling. Model fit, model comparison and regression diagnostics for each method are covered. In addition to these topics, students are also introduced to techniques for causal inference, imputation and reproducible research. All methods will be illustrated with an appropriate demographic survey data set and applied examples from real research.

Computer skills:

We will be using R. R is the language in which most research statisticians work and current methodological developments are being made, it is also free and can be used on any operating system.

Please update your R version (4.6.2) to the latest release and update all your packages.

We will use R through Positron (https://positron.posit.co/download), and we will use Posit Publisher to publish homework to Posit Connect Cloud. You turn in two links for each homework through Canvas: the published document and your GitHub repository.

I will also post data and other resources to my Github repository https://github.com/coreysparks/DEM7283

R packages for the class

Copy and paste the following command in Positron to install the packages we will use in this class:

source("https://raw.githubusercontent.com/coreysparks/DEM7283/master/install_first_7283.R")

In Positron. Please have these programs downloaded and installed prior to class, and create a free GitHub account. The setup page of the course book, https://coreysparks.github.io/DEM7283/setup.html, walks you through each step.

Revised calendar

Week	Topic
1	Course introduction; reproducible workflow with Quarto, renv, and Git
2	Complex survey design, weights, replicate variance
3	Binary outcomes: logit and probit, predicted probabilities, AMEs
4	Model comparison across nested logits (KHB); prediction and cross validation
5	Ordinal and multinomial models
6	Counts and rates: Poisson, negative binomial, offsets, zero inflation
7	Decomposition analysis: Kitagawa and Oaxaca-Blinder
8	Spring break (optional review in the course book)
9	Missing data and multiple imputation with survey designs
10	Longitudinal models: GEE, fixed effects, within-between models
11	Nonlinear regression: Splines and GAMs
12	Multilevel models I: centering, random slopes, level-specific weights
13	Multilevel models II: GLMMs, growth curves, APC identification
14	Causal inference foundations: Propensity score weighting
15	Tree-based machine learning with tidymodels
16	Final papers due